

NAME:SAMARTH CHAND THAKURI

SAP ID:500119430

BATCH:B1 CCVT

1. `package Experiment9;`

`import java.util.*;`

`public class PrimeCheck {`

`public static boolean isPrime(int num) {`

`if (num <= 1) return false;`

`for (int i = 2; i <= Math.sqrt(num); i++)`

`if (num % i == 0) return false;`

`return true;`

`}`

`public static void main(String[] args) {`

`ArrayList<Integer> numbers = new`

`ArrayList<>(Arrays.asList(2, 4, 5, 9, 13, 17));`

`for (Integer num : numbers) {`

`int value = num;`

`System.out.println(value + " is " + (isPrime(value)`

`? "Prime" : "Not Prime"));`

`}`

`}`

`}`

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2
2 is Prime
4 is Not Prime
5 is Prime
9 is Not Prime
13 is Prime
17 is Prime
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```

2. `package Experiment9;`

```
import java.util.*;

class Employee {
    String name;
    int id;
    double salary;

    Employee(String name, int id, double salary) {
        this.name = name;
        this.id = id;
        this.salary = salary;
    }
}
```

```
    public String toString() {
        return "Employee [Name=" + name + ", ID=" + id + ",
Salary=" + salary + "]\n";
    }
}

public class EmployeeManager {
    public static void main(String[] args) {
        ArrayList<Employee> employees = new ArrayList<>();

        employees.add(new Employee("Alice", 101, 50000));
        employees.add(new Employee("Bob", 102, 60000));
        employees.add(new Employee("Charlie", 103, 55000));

        // Update Bob's salary
        for (Employee emp : employees) {
            if (emp.id == 102) {
                emp.salary = 65000;
            }
        }

        // Remove Charlie
        employees.removeIf(emp -> emp.id == 103);

        // Print remaining
        for (Employee emp : employees) {
            System.out.println(emp);
        }
    }
}
```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe'
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2b\
Employee [Name=Alice, ID=101, Salary=50000.0]
Employee [Name=Bob, ID=102, Salary=65000.0]
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```

3. `package Experiment9;`

```
import java.util.*;

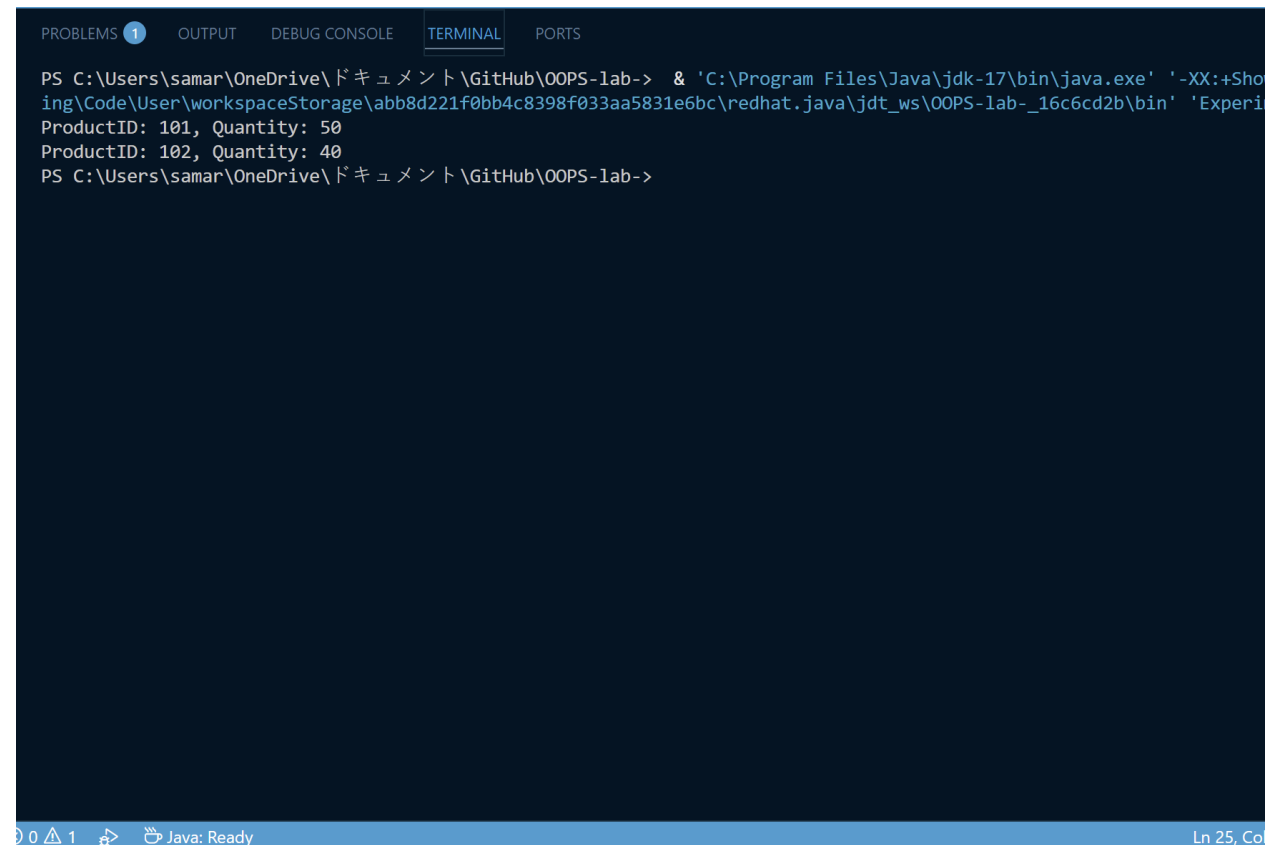
public class ProductInventory {
    public static void main(String[] args) {
        HashMap<Integer, Integer> inventory = new HashMap<>();

        inventory.put(101, 50);
        inventory.put(102, 30);
        inventory.put(103, 20);

        inventory.put(102, inventory.get(102) + 10);
    }
}
```

```
inventory.remove(103);

for (Map.Entry<Integer, Integer> entry :
inventory.entrySet()) {
    System.out.println("ProductID: " + entry.getKey() +
", Quantity: " + entry.getValue());
}
}
```



The screenshot shows an IDE interface with a terminal window. The terminal has tabs for PROBLEMS (1), OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, displaying the following text:

```
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+Sho
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2b\bin' 'Experi
ProductID: 101, Quantity: 50
ProductID: 102, Quantity: 40
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```

At the bottom of the IDE, there is a status bar with icons for a file explorer, a search icon, and a Java logo with the text "Java: Ready". On the right side of the status bar, it says "Ln 25, Co".

4. `package Experiment9;`

```
import java.util.*;

public class UniqueName {
    public static void main(String[] args) {
        String[] names = {"Aman", "Varchasv", "Sourabh",
"Varchasv", "Aman"};

        HashSet<String> nameSet = new
HashSet<>(Arrays.asList(names));

        String search = "Sourabh";
        System.out.println(search + " exists: " +
nameSet.contains(search));

        System.out.println("Unique names:");
        for (String name : nameSet) {
            System.out.println(name);
        }
    }
}
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\
mar\AppData\Roaming\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java
Sourabh exists: true
Unique names:
Varchasv
Aman
Sourabh
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```

0 1 Java: Ready

5. `package Experiment9;`

```
import java.util.*;

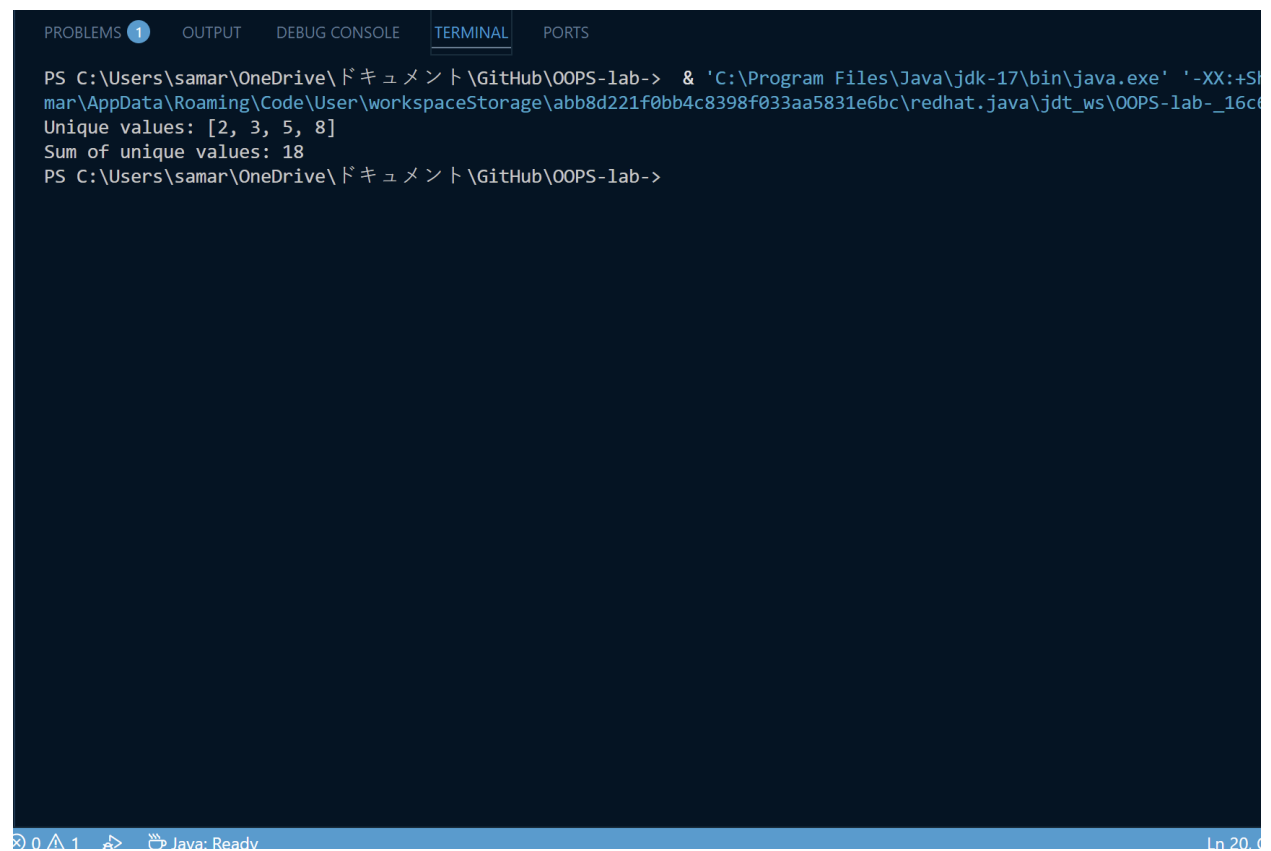
public class UniqueSum {
    public static void main(String[] args) {
        ArrayList<Integer> numbers = new
ArrayList<>(Arrays.asList(3, 5, 3, 8, 2, 5));

        HashSet<Integer> uniqueNumbers = new HashSet<>(numbers);

        int sum = 0;
```



```
        for (int num : uniqueNumbers) {  
            sum += num;  
        }  
  
        System.out.println("Unique values: " + uniqueNumbers);  
        System.out.println("Sum of unique values: " + sum);  
    }  
}
```



The screenshot shows an IDE interface with a terminal window. The terminal displays the output of a Java program. The command executed is `PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+Sh...`. The output is `Unique values: [2, 3, 5, 8]` and `Sum of unique values: 18`. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The status bar at the bottom shows 'Java: Ready' and 'Ln 20, C...

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+Sh  
mar\AppData\Roaming\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c  
Unique values: [2, 3, 5, 8]  
Sum of unique values: 18  
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->  
Java: Ready  
Ln 20, C...
```